

Week 4 Project Summary

HealthConnect Project Kickoff and Problem Understanding

Experience Lab Internship Programme

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1. Problem My Track Will Address

As the Data Science intern, my role in the HealthConnect project is to determine whether the available appointment data can support a machine learning solution that predicts the likelihood of a patient missing a scheduled appointment (a “no-show”), so the clinic can intervene proactively (e.g., targeted reminders, informed overbooking) rather than reacting after appointments are missed.

2. Resources Used

- Week 4 Assignment Brief (AnalystLab Africa HealthConnect Project Kickoff document).
- HealthConnect_Appointment_Data.csv — obtained and profiled (5,000 appointment records, 18 fields).
- HealthConnect_Data_Dictionary field descriptions — reviewed to confirm field meanings; full cross-check against the official file outstanding for Week 5.
- Jupyter notebook (HealthConnect_DS_Week4_EDA.ipynb) built and executed to run the data quality assessment and exploratory analysis in Python (pandas, matplotlib).

3. Key Observations from Week 4

- The business problem (reducing missed appointments) maps cleanly onto a binary classification ML problem: `appointment_outcome` provides a usable, clean target.
- Data quality is good — no duplicate records, no logical inconsistencies (e.g., `previous_no_shows` never exceeds `previous_appointments`, dates are always in the correct order), and only small (1–2%) genuine missing-value rates in distance and waiting time.
- The target is close to balanced (48.5% No-Show vs. 46.3% Attended vs. 5.3% Cancelled) — this was not assumed at the start of the week and simplifies later class-imbalance handling.
- **booking_lead_days** is the strongest predictor found: no-show rate rises from ~28% (0–7 days' notice) to ~68% (45–60 days' notice). **previous_no_shows** is the second strongest signal.
- Categorical fields (gender, appointment type, day/time, reminder channel) show only weak, flat relationships with the outcome individually.
- Each patient appears an average of ~3 times in the data, which means train/test splitting must be patient-grouped or chronological, not a random row split, to avoid leakage.
- **reminder_channel** missing values (27.3%) are structured and coincide exclusively with **reminder_sent** = “No”

4. Proposed Approach

- Treat this as a supervised binary classification problem with a no-show target (1 = missed appointment, 0 = attended), modelling Cancelled appointments (~5%) as a separate category.
- Use a patient-grouped or chronological train/validation/test split, since a random split risks leaking patient-specific behaviour and the model's real-world use is predicting future appointments.
- Start with an interpretable logistic regression baseline built on the two strongest features (booking_lead_days, previous_no_shows), then compare against tree-based ensemble models (Random Forest, Gradient Boosting).
- Evaluate with precision, recall, F1, and ROC-AUC, and use feature importance so results are explainable to clinic staff.

5. Key Considerations That May Affect the Project

- Leakage risk: waiting_time_minutes is only known once an appointment is underway and has been excluded from candidate features for this reason.
- Fairness and ethics: demographic fields showed weak standalone relationships with the outcome, which is reassuring, but combined-feature bias checks are still planned once a model is trained.
- Cross-track alignment: target/outcome definitions should be agreed with the Data Analytics track to avoid duplicated or conflicting analysis.
- Address potential data leakage, ensure robust imputation for missing distance data, and prioritize Recall to reduce missed operational interventions

6. Proposed Focus for Week 5

- Engineer the prioritised features (booking_lead_days, previous_no_show_rate) and implement a patient-grouped/chronological split.
- Build and evaluate a first baseline logistic regression model, then compare against ensemble methods.
- Extend the Week 4 notebook into the modelling phase and push updates, plus an updated README, to the GitHub repository.